

Instructions for homework timeline and submission

1. In this homework you will work within teams of 4 classmates. The team assignments are uploaded on CANVAS. You can initially reach out to your team members via CANVAS.
2. Each team is randomly assigned to one of the three studies. Please see assignments on CANVAS.
3. You will present your work with your team on **December 1 and 3** during class time (**3.35-4.50pm MT**). The assigned time slot for your team has been announced on CANVAS. Most of the work should be ready by the class presentations, including the data analysis and the presentation. **Participation in the presentation on both days (December 1 and 3) is mandatory will count toward the in-class participation grade. Everyone will be called on by name to ask questions during the presentations.** Each team will have **4 minutes to present**, followed by **4 minutes of questions**.
4. The final report is due on **December 4, 2025 @ 11.59pm**. Please create a zip file with two pdf files, including the final report and the presentation. **One member per team can make the submission.**
5. You can use any publicly available library or code for this homework.
6. The total for this homework is **14 points** (out of 100 total for the class) and **2 Bonus points**.

Scope of Work

Speech-based machine learning (ML) technologies are transforming digital health and education by enabling systems to process and understand human cognition, affect, and interpersonal interactions. In digital mental health, these technologies assist in diagnosing, monitoring patient treatment by analyzing speech data. In education, these systems can serve as a foundation to intelligent tutoring systems that can track the learners' responses, assess performance, and provide personalized feedback.

This homework demonstrates the use of these systems in mental health and job interview training via three studies. It further examines ways to make these technologies more responsible via increasing their explainability and enhancing data privacy.

Study 1: Designing explainable speech-based machine learning models of depression

Patients with depression depict differences in language use and prosodic patterns compared to their healthy counterparts, such as an increased use of first-person singular pronoun use, increased use of negative emotion words, and decreased use of positive emotion words [8]. In addition, depression affects psychomotor control and the phonological loop [5], leading to changes in speech articulation, prosody, and vowel characteristics, such as shorter duration and reduced vowel space [2]. This study explores the effectiveness of speech-based ML models for automatically estimating the degree of depression severity. It further investigates the use of explainability methods to provide insights into these models, with the goal of informing and supporting clinical practice.

The data for this study comes from the Extended Distress Analysis Interview Corpus (E-DAIC) dataset [4], which contains clinical interviews designed to support the diagnosis of psychological distress conditions such as depression. More details about the dataset, including the experimental setup and type of data, can be found here: <https://dcapswoz.ict.usc.edu/>. The data uploaded on CANVAS includes interview transcripts from 190 participants. The ‘DepressionLabels.xlsx’ file contains information on participants’ depression severity (PHQ_Score). The folder ‘E-DAIC_Transcripts’ includes csv files (one per participant) with the transcript data. Each filename, named as ‘x_Transcript.csv’, where x is the participant ID, includes the transcripts from the responses provided by participant x during the clinical interview. Please disregard any additional depression labels in the labels file that do not correspond to participants for whom transcript files are available. The folder ‘E-DAIC_Acoustics’ includes csv files with acoustic features. Each csv file includes acoustic features for one speaker extracted at the turn-level. Column 1 of the file has the participant ID (PID), columns 3-4 include the start and end time of each turn, column 5 includes the text corresponding to the turn, and columns 7 and onward include the acoustic measures that were extracted with the Extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) [3].

Randomly split the participants into 5 folds and report results accordingly in the following experiments.

(a) (2 points) Extracting language features. Extract several language features from the data. Include at least two of the following types of features, spanning different levels of complexity:

- Syntactic vectorizers: count vectorizer (e.g., *CountVectorizer* from sklearn) transforming a collection of text documents into a numerical matrix of word or token counts; TF-IDF vectorizer (e.g., *TfidfVectorizer* from sklearn) incorporating document-level weighting, which emphasizes words significant to specific documents’ part-of-speech features counting the distribution of part of speech tags over a document
- Semantic features: sentiment scores (e.g., Vader, <https://github.com/cjhutto/vaderSentiment>), topic distribution (using topic modeling), or named entities
- Advanced features: word embeddings, such as Word2Vec or BERT (e.g., *pytorch-pretrained-bert*) for capturing contextual meaning

Below are some additional resources that could be useful for feature extraction: NLTK toolkit (<https://www.nltk.org/>); Google word2vec (<https://code.google.com/archive/p/word2vec/>);

Hugging Face (<https://huggingface.co/>); Jurafski & Martin, Speech & Language Processing, Chapter 6: Vector Semantics and Embeddings (<https://web.stanford.edu/~jurafsky/slp3/6.pdf>).

(b) (2 points) Estimating depression severity with interpretable models using language features. Use a decision tree to estimate the degree of depression for each participant based on the language features extracted in question (a). Please report the Pearson's correlation r and absolute relative error (RE) between the estimated and actual PHQ-8 scores for all participants included. Experiment with different values of k . Please discuss your findings (e.g., What are the most informative language features of depression based on the first levels of the decision tree and how can they be interpreted?).

Note: The absolute relative error, RE , is defined as follows:

$$RE = \frac{|\text{estimated PHQ-8} - \text{actual PHQ-8}|}{\max(\text{PHQ-8})}$$

(c) (2 points) Estimating depression severity with interpretable models using acoustic features. Use a decision tree to estimate the degree of depression for each participant based on the provided acoustic features. Please report the Pearson's correlation r and absolute relative error (RE) between the estimated and actual PHQ-8 scores for all participants included. Experiment with different values of k . Please discuss your findings (e.g., What are the most informative acoustic features of depression based on the first levels of the decision tree and how can they be interpreted?).

Note: You can find more information on the acoustic features, included in the Extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS), in this link: <https://sail.usc.edu/publications/files/eyben-preprinttaffc-2015.pdf>.

(d) (2 points) Estimating depression severity with unimodal and multimodal deep learning models. Use a deep learning ML algorithm of your choice (e.g., LSTMs) to estimate the degree of depression for each participant. Conduct experiments using language and acoustic features separately as two distinct modalities, as well as using their combination. Please report the Pearson's correlation r and absolute relative error (RE) between the estimated and actual PHQ-8 scores for all participants included. Please discuss your findings (e.g., How does the performance of the unimodal deep learning model compare to that of decision trees? How did different hyper-parameters affect the model's performance? How do the two modalities compare to one another? How does the performance of the multimodal model compared to the unimodal ones?).

(e) (2 points) Explainable ML. Use an explainable ML algorithm for interpreting the decisions and decision-making process of the deep learning models from question (d). Discuss the pros and cons of each ML algorithm and the corresponding types of explanations that are provided. You can evaluate the provided explanations in terms of various criteria such as comprehensibility, relevance, and scalability.

Note: You can try different algorithms, such as the EBM-Explainable Boosting Machine (<https://interpret.ml/docs/ebm.html>), the SHAP-SHapley Additive exPlanations (<https://shap.readthedocs.io/en/latest/>), LIME-Local Interpretable Model-Agnostic Explanations (<https://github.com/marcotcr/lime>), etc. Please see [1], uploaded under filename 'explainabil-

ity.pdf' for additional discussion and resources on ML explainability.

(f) (Bonus, 2 points) Experimenting with transformers. Use a pre-trained transformer-based model (e.g., quantized Llama, miniGPT) to conduct depression severity estimation based on the provided transcripts. Experiment with prompt engineering or task-specific prompts to guide the model in adapting to each classification objective, including incorporating a few labeled example transcripts within the prompt. Please use the same evaluation metrics as in **(d)**. How does the performance of this model compare to the previous models? Please provide a discussion for all participants and for different participant groups.

Note: You can find a useful github repo here: <https://github.com/karpathy/minGPT>

(g) (2 points) Presentation. Create a presentation of your work. The presentation will provide the main gist of your work, including the problem statement, your methodology, and the main results from your experiments. **Add visuals so that people understand the main concepts.** Each team will have 4 minutes to present, followed by 4 minutes of questions.

Note: Each team will present in class on **December 1 and 3** during class time (**3.35-4.50pm MT**). The assigned time slot for your team has been announced on CANVAS. **Participation in the presentation for both days (December 1 and 3) is mandatory and will count toward the in-class participation grade. Members from other teams will be called on by name to ask questions during the presentations.**

(h) (2 points) Teamwork. Please follow the survey link below, in which you will need to discuss the level of engagement of your team members. The feedback that your team members will provide for one another will be used as an indication of each member's contribution to this group assignment.

Survey Link: <https://forms.gle/VcmzedyqTrnQ9Fre7>